DiagBeacon

DNS-Based Discovery of Diagnostic Services

Discover the right endpoint. Verify it safely. Explain what failed.

TEAM NETSPHERE

Problem Statement PS-033

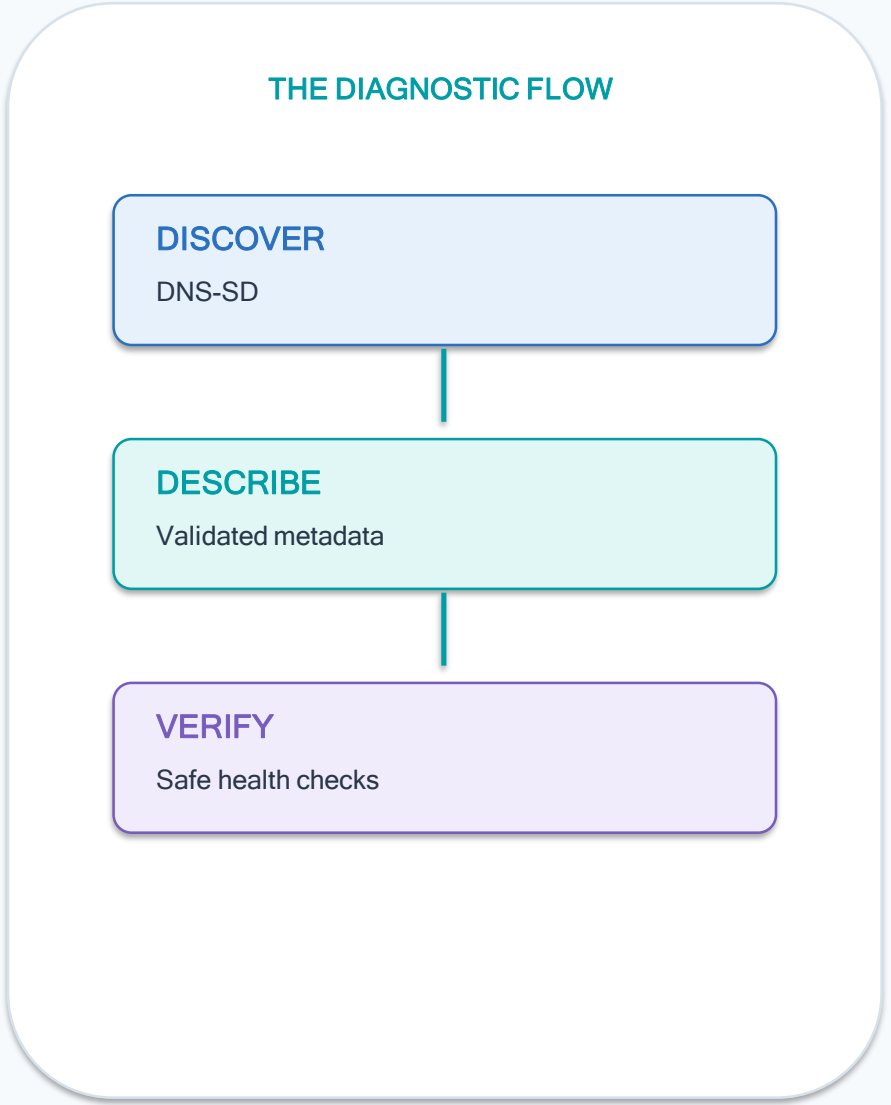
TEAM MEMBERS

Ritika • Anil Chaure

FACULTY MENTOR

Dr. Vijay Dhawale

DY Patil



Team code: A3-PS033-TC232 • A lightweight, evidence-led diagnostic layer for authorized environments

From service discovery to a clear next action

DiagBeacon connects discovery, validation and safe checks in one operator workflow.

01 DISCOVER

Find advertised diagnostic services through DNS-SD records.

02 VALIDATE

Check metadata, schema, endpoint, version and auth requirements.

03 VERIFY SAFELY

Run only allowlisted, rate-limited DNS / TCP / TLS / HTTP(S) checks.

04 EXPLAIN

Show evidence, freshness, status and a practical next step.

EXAMPLE OUTPUT

REACHABLE

AUTH REQUIRED

STALE

UNREACHABLE

Deterministic checks provide the evidence; AI-assisted wording is optional, not the source of truth.

It turns “I found a service” into “I know what to do next”

The real gap is not discovery alone – it is the missing context between discovery and diagnosis.

Scattered context

Service details are split across manuals, dashboards, configuration and logs.

DIAGBEACON **One validated service catalogue**

Moving targets

Hostnames, ports and DNS records change; static lists become stale.

DIAGBEACON **Freshness checks and rediscovery**

Ambiguous status

A listed endpoint may be unreachable, auth-protected or unhealthy.

DIAGBEACON **Layered checks with evidence**

Manual guesswork

Operators repeat steps and infer causes from incomplete signals.

DIAGBEACON **Consistent results and next steps**

Expected value: less time searching, fewer false assumptions, safer checks and a traceable diagnostic history.

We build on existing standards — then connect the missing steps

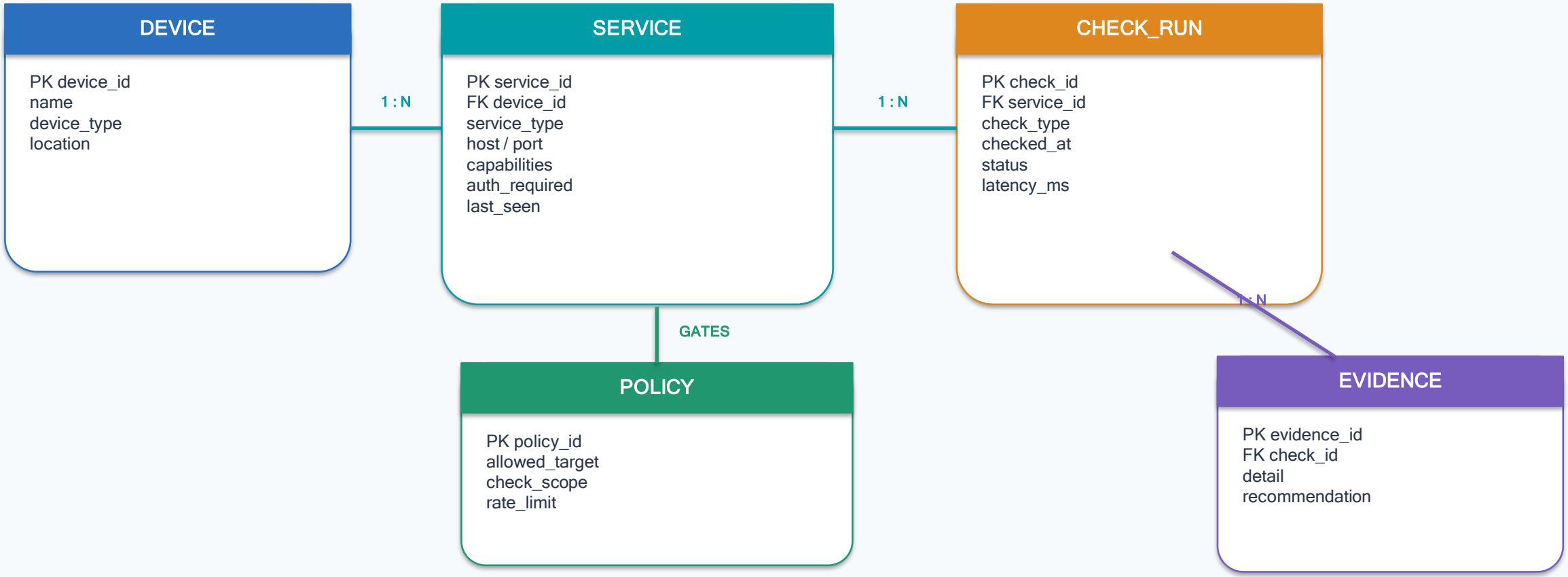
Position DiagBeacon as a lightweight diagnostic layer, not a replacement for established discovery tools.

EXISTING APPROACHES		PROPOSED DIAGBEACON WORKFLOW	
DNS-SD / Bonjour / Avahi	Advertise and discover services	Unified diagnostic catalogue	Discovery results plus capability metadata
Consul	Service registry and configured health checks	Schema + freshness validation	Flag malformed or outdated service records
upwatch	DNS-SD discovery plus HTTP(S) uptime checks	Policy-gated layered checks	Only approved endpoints; rate limits and timeouts
Manual correlation	Operator joins logs, endpoints and context by hand	Evidence-led explanation	Status, reason, history and recommended next action

Differentiator: an integrated, safe and explainable workflow – not a claim that the underlying protocols are new.

ER diagram: keep services, checks and evidence connected

A compact relational model supports history, traceability and exports.



Keys and relationships are proposed for the MVP; SQLite is sufficient for a local prototype.

DiagBeacon

Simplified Working Architecture

End-to-end flow from service discovery to diagnostics and visualization

Policy & Security

- Allowlist / Denylist
- Rate limits & Timeouts
- Only approved endpoints

1 IoT / Diagnontic Devices



IP Camera



Sensor



PLC / Controller



Medical Device

Devices advertise diagnostic services using DNS-SD records (PTR / SRV / TXT)

2 Discovery Engine



- Queries DNS-SD records
- Parses PTR / SRV / TXT
- Resolves endpoints

3 Diagnostic Engine



- DNS Resolution
- TCP Connectivity
- HTTP(S) Health Check
- Authentication Check

4 Analysis & Result



- Classifies status
- Finds failure reason
- Shows evidence
- Suggests next steps

5 Dashboard



- Service inventory
- Real-time status
- Detailed results & history
- Export (CSV / JSON)

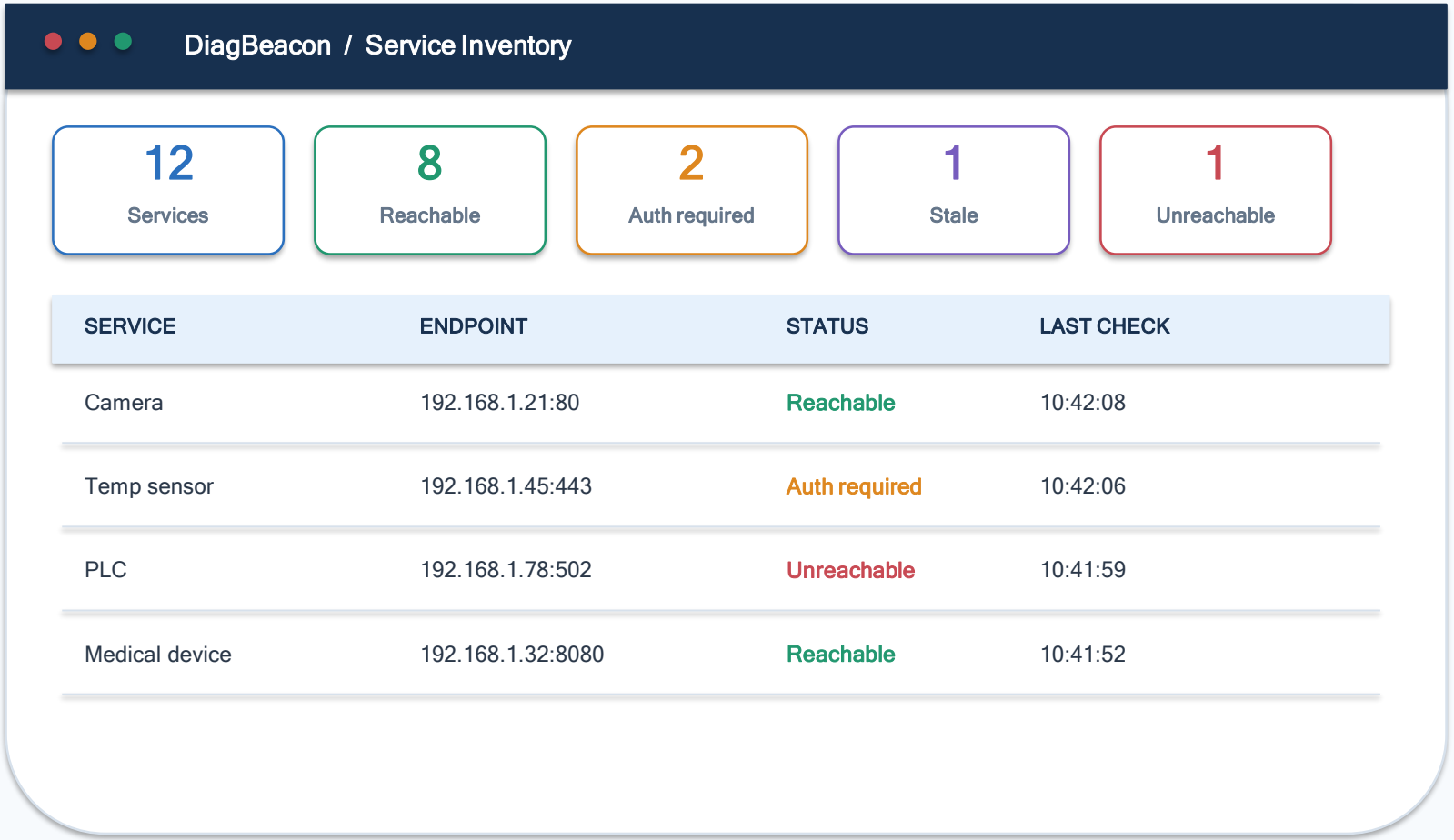
Database (SQLite)



- Service inventory
- Check results
- Historical data
- Evidence logs

Prototype Dashboard: Service Inventory

A unified view of discovered services, endpoint status, and recent diagnostic results.



PROTOTYPE WORKFLOW

1. Discover

Load DNS-SD records or seeded demo data.

2. Validate

Show schema and allowlist result.

3. Run checks

Display DNS / TCP / TLS / HTTP(S) evidence.

4. Explain

Show status, reason, timestamp and next step.

Illustrative dashboard with sample data; live checks will be connected during MVP implementation.

Standards first. Clear boundaries. Verifiable foundations.

These sources support the protocols and tools used in the proposed design.

RFC 6763 – DNS-Based Service Discovery

Protocol specification for DNS-SD.

https://www.rfc-editor.org/rfc/rfc6763.html

Avahi

Open-source implementation of mDNS / DNS-SD.

https://www.avahi.org/

Consul – Service Discovery

Reference for service registry and discovery concepts.

https://developer.hashicorp.com/consul/docs/discover

Consul – Health Checks

Reference for configured health-check patterns.

https://developer.hashicorp.com/consul/docs/reference/service/health-check

upwatch repository

Existing DNS-SD discovery and uptime-checking project.

https://github.com/jasonrogena/upwatch

AIORI-3 Hackathon Guidelines

Competition context and submission guidance.

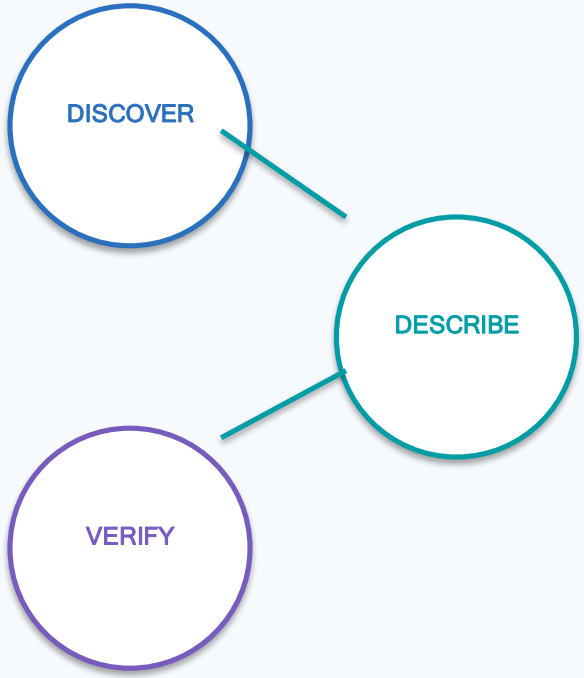
https://portal.aiori.in/aiori-3/hackathon-guideline/

Differentiation and prototype features are proposed design choices to be validated during implementation and testing.

Thank you.

Discover. Describe. Verify safely.

DiagBeacon turns scattered service details into clear, evidence-led next steps.



TEAM NETSPHERE • A3-PS033-TC232

Ritika | Anil Chaure | Dr. Vijay Dhawale

Questions? We welcome feedback on the safety model, data design and evaluation plan.